

6)

$$v_i = \frac{e_i \cdot e_k}{\sum_j e_j \cdot e_k}$$

$$v_i \cdot e_j = 0$$

$$v_i \cdot e_k = 1$$

$$v_i \cdot e_i = \frac{e_i \cdot e_i}{\sum_j e_j \cdot e_i} = 1$$

$$v_i \cdot e_L = \delta_{iL}$$

$$e_i = \frac{e_i \cdot e_k}{\sum_j e_j \cdot e_k}$$

se normaliza a e_i

$$e_i \cdot e_i = \delta_{ii}$$

$$(a \times b) \times (c \times d)$$

$$\epsilon^{ijk} (a_k b_j) \epsilon^{lmn} (c_n d_m)$$

$$\epsilon^{ijk} \epsilon^{lmn} (a_k b_j \times c_n d_m)$$

$$\epsilon^{ijk} \epsilon_{ilm} a^l b^m \epsilon_{knp} c^p d^p$$

$$\epsilon^{kij} \epsilon_{lmj} = (\delta_l^k \delta_m^i - \delta_l^i \delta_m^k)$$

$$(\delta_l^k \delta_m^i) \epsilon_{knp} a^l b^m c^p d^p - (\delta_l^i \delta_m^k) \epsilon_{knp} a^l b^m c^p d^p$$

$$b^m [\epsilon_{knp} a^k c^p d^p] - a^l [\epsilon_{knp} b^m c^p d^p]$$

$$b(a \cdot (c \times d)) - a(b \cdot (c \times d))$$

$$a = e_1 = d \quad b = e_2 \quad c = e_3$$

$$e_2 (e_1 \cdot (e_3 \times e_1)) - e_1 (e_2 \cdot (e_3 \times e_1))$$

$$-e_1 v$$

$$\tilde{v} = -\frac{1}{v^3} v (e_2 \times e_3) \cdot e_1 = 1 \frac{v^2}{v^3} = \frac{1}{v}$$

$$v \tilde{v} = v \frac{1}{v} = 1$$

c) $a \cdot e^i = 1$ supongo $b \cdot e^i = 1$

$$a \cdot e^i = b \cdot e^i \rightarrow (a - b) e^i = 0$$

$$a = b \rightarrow a \neq \text{unka}, a \neq 0$$

$$a = e_1 + e_2 + e_3$$

$$(e_1 + e_2 + e_3) \cdot e^i = e_1 e^i + e_2 e^i + e_3 e^i = \delta_1^i + \delta_2^i + \delta_3^i$$

$$w_1 = (4, 2, 1) \quad w_2 = (3, 3, 0) \quad w_3 = (0, 0, 2)$$

$$v = w_1(w_2 \times w_3) = w_2(w_3 \times w_1) = w_3(w_1 \times w_2)$$

$$w_2 \times w_3 = [6, -6, 0]; \quad w_3 \times w_1 = [-4, 8, 0]; \quad w_1 \times w_2 = [-3, 3, 6]$$

$$v = 12$$

$$e^1 = \frac{1}{12} [6, -6, 0]; \quad e^2 = \frac{1}{12} [-4, 8, 0]; \quad e^3 = \frac{1}{12} [-3, 3, 6]$$

$$e^i w_j = \delta_j^i$$

contra vortices

$$a = [1, 2, 3]$$

$$\rightarrow a^i = e^i a$$

$$\frac{1}{12} a^1 = \left[\frac{1}{2}, -\frac{1}{2}, 0 \right] \cdot [1, 2, 3] = -\frac{1}{2}$$

$$a^2 = \left[-\frac{1}{3}, \frac{2}{3}, 0 \right] \cdot [1, 2, 3] = 1$$

$$a^3 = \left[-\frac{1}{4}, \frac{1}{4}, \frac{1}{2} \right] \cdot [1, 2, 3] = \frac{3}{4}$$

$$a^1 w_1 + a^2 w_2 + a^3 w_3 = [1, 2, 3]$$

co vortices

$$a_i = w_i a$$

$$a_1 = [4, 2, 1] \cdot [1, 2, 3] = 11; \quad a_2 = [3, 3, 0] \cdot [1, 2, 3] = 9$$

$$a_3 = [0, 0, 2] \cdot [1, 2, 3] = 6$$

$$a_1 e^1 + a_2 e^2 + a_3 e^3 = [1, 2, 3] \quad \leftarrow 11 \cdot \frac{1}{12} + 9 \cdot \frac{1}{12} + 6 \cdot \frac{1}{12} = 1$$

$$a = a^i w_i \quad 0 \leq b^j w_j; \quad a \cdot w_i = 0 \quad \leftarrow a \cdot w_i = 0$$

$$a^i w_i \times b^j w_j = (a^1 w_1 + a^2 w_2 + a^3 w_3) \times (b^1 w_1 + b^2 w_2 + b^3 w_3)$$

$$i = j \Rightarrow 0$$

$$a^1 b^2 (w_1 \times w_2) + a^1 b^3 (w_1 \times w_3) + a^2 b^1 (w_2 \times w_1) + a^2 b^3 (w_2 \times w_3) + a^3 b^1 (w_3 \times w_1) + a^3 b^2 (w_3 \times w_2)$$

$$(a^1 b^2 - a^2 b^1) \underbrace{(w_1 \times w_2)}_{12 e^3} + (a^3 b^1 - a^1 b^3) \underbrace{(w_3 \times w_1)}_{13 e^2} + (a^2 b^3 - a^3 b^2) \underbrace{(w_2 \times w_3)}_{23 e^1}$$

$$12 [(a^1 b^2 - a^2 b^1) e^3 + (a^3 b^1 - a^1 b^3) e^2 + (a^2 b^3 - a^3 b^2) e^1]$$

$$12 \epsilon_{ijk} a^i b^j e^k$$

$$A \times b = 12 \epsilon_{ijk} a^i b^j e^k$$

± 7)

Mapa de Pauli

$$\begin{pmatrix} a & a + bi \\ a - bi & c \end{pmatrix}$$

$$\sigma_0 = I ; \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} ; \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} ; \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\text{Tr}(\sigma_i \sigma_j) = 2 \delta_{ij}$$

$$\tilde{\sigma}_i = \frac{1}{2} \sigma_i \quad \rightarrow \langle \sigma_i, \sigma_j \rangle = \text{Tr}(\tilde{\sigma}_i^\dagger \sigma_j) \rightarrow$$

$$\sigma_i^\dagger = \frac{1}{2} \sigma_i^\dagger = \frac{1}{2} \sigma_i = \tilde{\sigma}_i$$

$$\text{Tr}(\sigma_i, \sigma_i) \rightarrow \text{Tr}(\frac{1}{2} \sigma_i, \sigma_i) = 2$$